MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 8798

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed tunable frequency range 2.7 to 2.9 GHz, rated peak power output 800 kW, external magnet, coaxial line output, forced-air cooled.

ABSOLUTE RATINGS:

Parameters:	Ef	epy	ib	Pi	pi	tp
Units:	V	kv	a	W	kW	µsec
Maximum:	17	32	70	1300	200	2.5
Minimum:	--	--	--	--	--	---
	Note 1					

Parameters:	Du	tk	VSWR	Anode T	rrv	Alt.
Units:	--	sec	--	°C	kV/µs	ft
Maximum:	.0011	---	1.35	120	150	12,000
Minimum:	00	120	--	---	---	----

PHYSICAL CHARACTERISTICS:

Dimensions:	See outline drawing
Weight:	6.5 pounds approximately
Mounting position:	Any
Cathode:	Unipotential

TEST CONDITIONS:

Parameter:	Ef	Magnetic field	tpc	Du	rrv	prr	tk
Unit:	V	H	µsec	--	kV/µs	---	sec
Test 1:	---	2700	0.7	.0007	110	1000	180
Test 2:	---	2700	2.0	.0007	110	350	180
Test 3:	10	2700	1.0	.001	150	1050	---
Tolerance	±5%	±50	±0.1	--	---	---	---
Notes	Note 1	Note 4					

Parameter:	tfc	trc	trv	VSWR	Ib
Unit:	µsec	µsec	µsec	----	mAdc
Test 1:	0.50	0.125	0.15	1.15 max.	46
Test 2:	0.50	0.125	0.15	1.15 max.	46
Test 3:	0.50	0.125	0.15	1.35 max.	39.5
Tolerance	±0.10	±0.045	±0.05	----	±5%

GENERAL:

Qualification - Required

8798

FSC 5960

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification						
1031	High frequency vibration	--	No voltages See note 7	---	---	---	---
4004	Temperature coefficient	1	F1, F2 and F3 T = 60° to 90°C	ΔF/ΔT	---	70	kHz/A
1026	Low temperature	--	tk = 180 sec.	MP	---	1	%
	Quality conformance inspection, part 1						
D-30(b)	Dimensions	--	See figure 1	---	---	---	---
4003	Pressurizing	--	42.5 ±2.5 lb_f/in^2	---	---	---	---
4289	Heater current nonoperating	--	Ef = 16.0 V ±5% tk = 300 sec (min) See note 1	If	2.6	3.1	A
4306	Pulse voltage	1	F1, F3	epy	26	32	kv
4250	Power output (1)	1	F2, F3	Po	525	---	W
4250	Power output (2)	2	F1	Po	525	---	W
4308	R. F. bandwidth	1	F1, F2, F3	BW	---	2.5/ tpc	MHz
4308	Minor lobes	1	F1, F2, F3	Ratio	8	---	dB
4315	Stability	1	F1, F2, F3	MP	---	0.5	%
4223	Mechanical tuning range	1 1	Low frequency High frequency	F1 F3	--- 2.9	2.7 ---	GHz GHz
4250	Power output (3)	3	F1 and F3	Po	470	---	W
4315	Stability	2	See note 6	MP	---	0.5	%
	Quality conformance inspection, part 2						
4311	Pushing factor	3	F2	Ratio	---	125	kHz/A
4310	Pulling factor	1	I_b = 28 to 46 mAdc; F2	F	---	15.0	MHz
4223	Operating torque or force	--	See note 12/	Torque	---	20	oz-in
D-30(b)	Dimensions	--	See figure 1	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3						
----	Life test	1	Group S; F2	t	1000	---	hrs
----	Life test end points						
4250	Power output	1	F1, F2, and F3	Po	420	---	W
4308	R.F. bandwidth	1	F1, F2, and F3	BW	---	3.0/ tpc	MHz
4315	Stability	1	F1, F2, and F3	MP	---	1.0	%
4308	Minor lobes	1	F1, F2, and F3	Ratio	7	---	dB

NOTES:

1. Standby or preheat heater voltage shall be 16.0 Vdc +5%. During high voltage operation, heater voltage should be adjusted to the following approximate values.

Pi (watts)	Ef (Vdc)
1200-1400	5.5
1000-1200	8
800-1000	10.5
600-800	13
400-600	15
0-400	16

2. These periodic check tests which have been performed and have met specified requirements within 36 months need not be performed if test requirements, material or manufacturing processes have not changed in the interim.

3. The value of VSWR specified excludes the tube coupling adapter.

4. The magnetic field should be calibrated in accordance with the following procedure: See figure 4.

   a. With a conventional 1/8 inch pole piece attached to the pole face of the magnet opposite the magnetron tuner, the magnetic gap should be 1.800 inches.

   b. The magnetic field should than be adjusted for 2700 gauss at the center of the gap.

   c. Remove the conventional 1/8 inch pole piece and replace it with the distortion pole piece.

5. Frequency definitions:

   F1 = 2.7 GHz F2 = 2.8 GHz F3 = 219 GHz

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NOTES: - Continued

6. Missing pulses shall be counted during the last three minutes of a stability test interval not to exceed 6 minutes. The stability test interval shall first be conducted at F1 and shall be started no later than 10 minutes after the application of high voltage. Test intervals shall then be conducted consecutively at F2 and F3.

7. Following this test, the magnetron shall be capable of normal operation under Test Condition 1.

Custodian:
Air Force - 85

Review activity:
Air Force - 99

User activity:
Air Force - 11

Preparing activity:
Air Force - 85

(Project No. 5960-3060)

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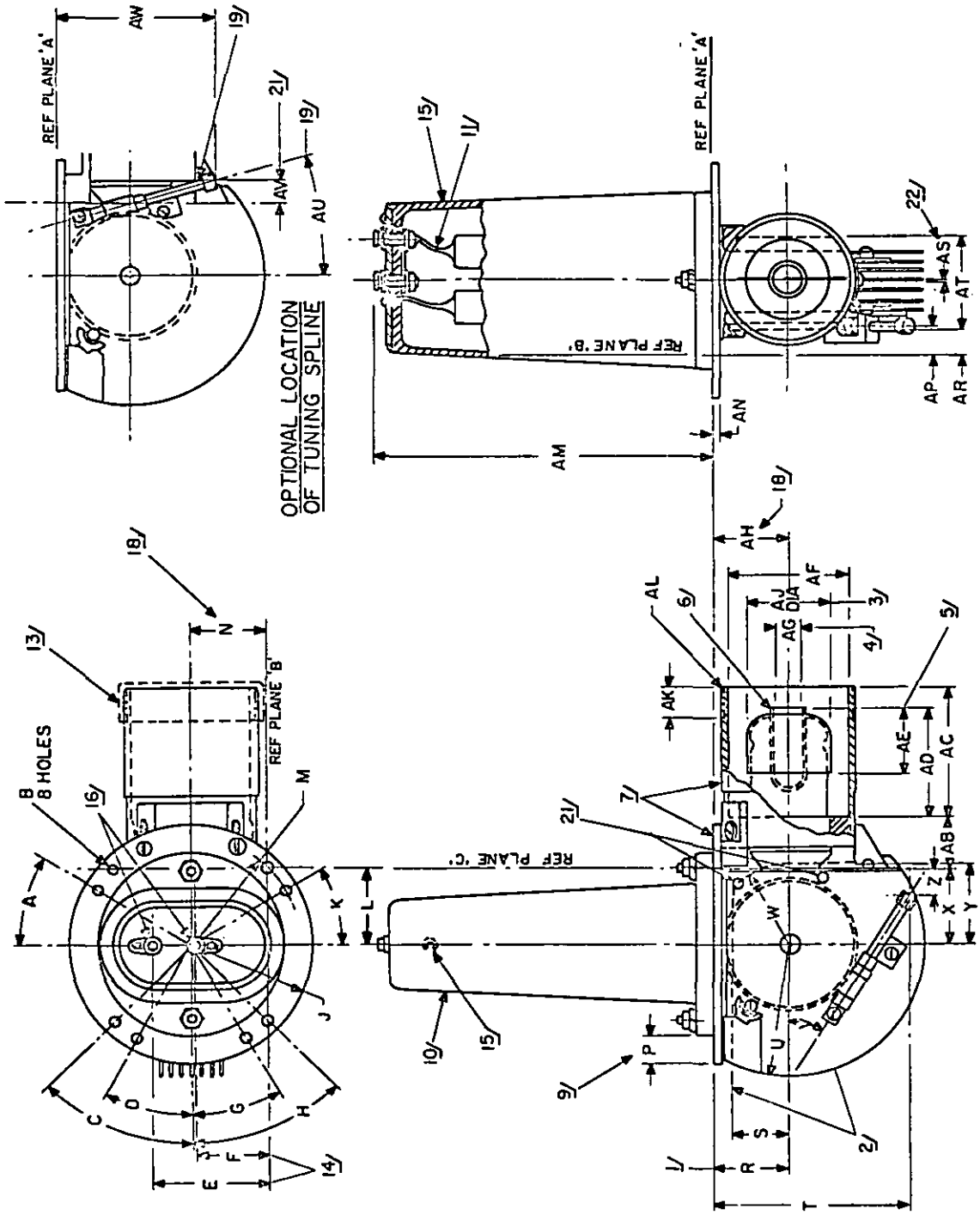


FIGURE 1 Outline drawing of electron tube type 8798

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OUTLINE DIMENSIONS SIMILAR TO 242 JAN (REF)

LTR	DIMENSIONS			LTR	DIMENSIONS		
	NOM	MIN	MAX		NOM	MIN	MAX
+ A		29°48'	30°12'	+ Y		1.500	
+ B		.205	.215	Z	.756		
+ C		44°48'	45°12'	+ AB		.803	.833
+ D		29°48'	30°12'	AC		2.287	2.307
E	2.156			AD		2.060	2.110
F	1.359			AE		1.125	
+ G		29°48'	30°12'	AF		2.314	2.328
+ H		44°48'	45°12'	AG		.550	.560
+ J		2.266R	2.296R	AH		1.420	1.460
+ K		29°48'	30°12	AJ			1.620
L	1.437			+ AK		.593	
+ M		2.029 R	2.035 R	AL	2-1/2-18 NS-2A		
N		1.417	1.457	+ AM		6.219	6.407
+ P		.500		AN	.187		
R	1.440			AP		.438	.688
+ S		1.063		AR		.525	.625
T	3.500			AS	.025		
+ U			2.656R	AT			1.740
+ V		30°	35°	AU		11°30'	16°30'
+ W		1.500R		AV	.313		
X	1.437			AW	2.812		

+ QUALITY CONFORMANCE INSPECTION, PART 2
OTHER DIMENSIONS QUALITY CONFORMANCE INSPECTION PART I

NOTES (MECHANICAL REQUIREMENTS)

1/ The periphery of the anode shall lie within a 2.160 dia circle located as specified for non-tunable side of anode.

2/ Maximum width specified applies to area defined by broken line and circumference of radiator.

3/ Center-line of maximum diameter shall be concentric with center-line of guard pipe to within .040.

4/ Applies to inner conductor insert only center-line of inner conductor insert shall be concentric with center of guard pipe to within .025.

5/ Applies to straight portion of inner conductor wall.

6/ No sharp edges of outside diameter at end of inner conductor.

7/ Tube may be supported by plate (mounting) or guard pipe.

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NOTES (MECHANICAL REQUIREMENTS) (Cont'd)

8/ Spline for adjusting tuning mechanism is as follows - 12 teeth 48 pitch - .250 pitch diameter.

9/ This annular area shall be flat within .015 (A thickness gage .125 wide shall not enter more than .250).

10/ Pyrex glass or approved equivalent.

11/ Leads shall be flexible and slack.

12/ Tuning mechanism will provide full range of tuning with five 360° turns, maximum.

13/ Protective guard for shipping purposes.

14/ The center of the jack holes shall be within a radius of .100 of the location specified but shall be spaced .797 ± .015 with respect to each other.

15/ Common cathode connection marked with letter "C".

16/ Hex locking head banana pin jack 19/32 long. hole .169 ± .005.

17/ Paint with heat resistant non-corrosive paint; the following shall be free from paint-top surface of mounting plate parts above mounting plate, screw threads on guard pipe and all surfaces inside guard pipe, tuning gear, stop assembly, and worm shaft assembly.

18/ Applies to location of center of guard pipe only.

19/ Clearance to adapter guard pipe must be sufficient to allow use of S. S. White Co. No. 2666X end fitting (13/32 diameter).

20/ All solder joints on mounting plate and guard pipe shall be soldered to provide an hermetic seal.

21/ Optional location of tuning spline, tube to be supplied with spline located as specified by customer.

22/ This dimension shows relation between a plane passing thru lateral center of anode and a plane thru center of guard pipe.

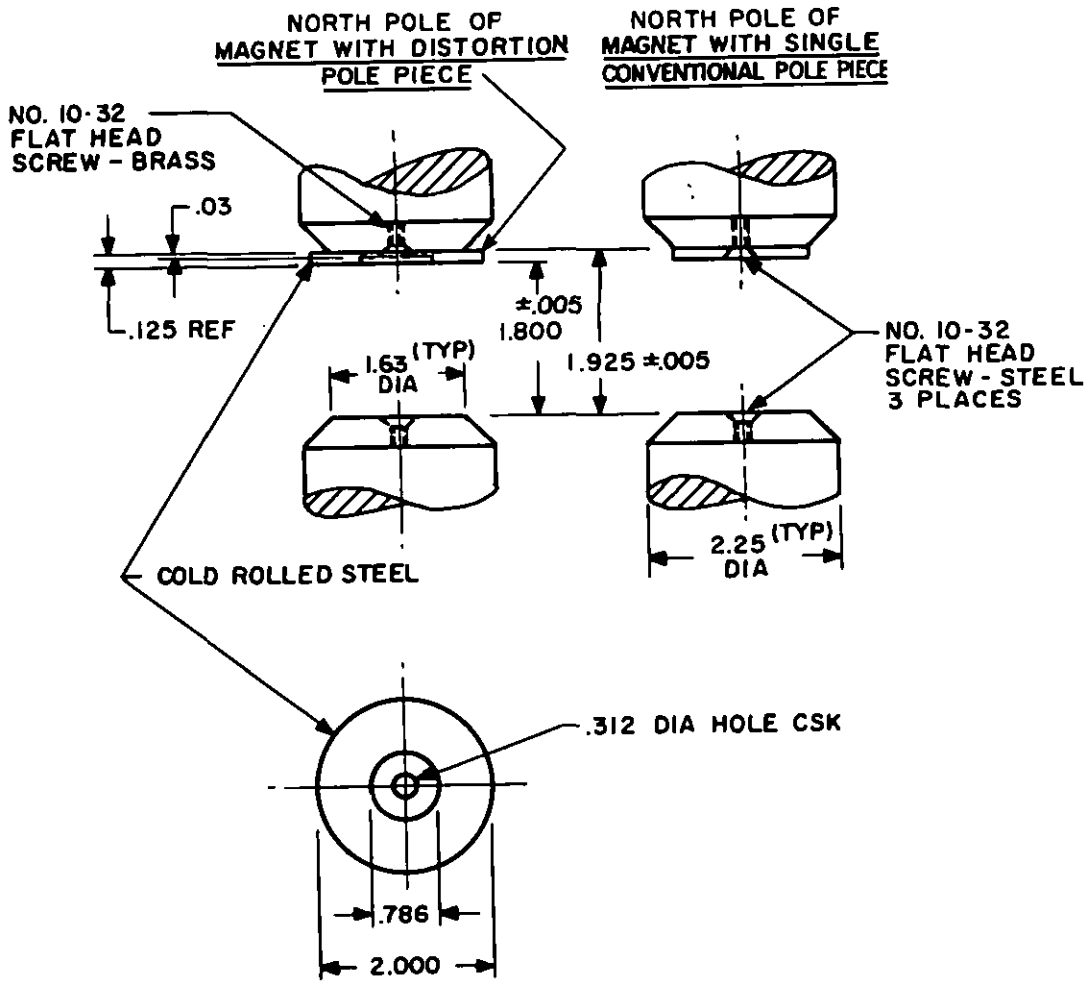
23/ Reference plane "A" is defined as a plane passing along the face of the mounting plate.

24/ Reference plane "B" is defined as a plane perpendicular to plane "A" passing through the center of the holes as shown.

25/ Reference plane "C" is defined as a plane mutually perpendicular to planes "A" and "B" passing through the center of the hole as shown.

26/ The letter "A" has been added to indicate external thread, not unified thread.

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCES ±.03

FIGURE 4 Magnetic Field Calibrators for Electron Tube Type 8798

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